

Deep Learning Practical No.1

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Roll No.24

```
import pandas as pd
import numpy as np
```

```
data_url = "http://lib.stat.cmu.edu/datasets/boston"
raw_df = pd.read_csv(data_url, sep="\s+", skiprows=22, header=None)
```

```
data = np.hstack([raw_df.values[::2, :], raw_df.values[1::2, :2]])
target = raw_df.values[1::2, 2]
```

```
columns = [
    "CRIM", "ZN", "INDUS", "CHAS", "NOX", "RM",
    "AGE", "DIS", "RAD", "TAX", "PTRATIO",
    "B", "LSTAT"
]
```

```
df = pd.DataFrame(data, columns=columns)
df['MEDV'] = target
```

```
print(df.head(10))
```

```
<>:2: SyntaxWarning: invalid escape sequence '\s'
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/tmp/ipykernel_663/159347055.py:2: SyntaxWarning: invalid escape sequence '\s'
raw_df = pd.read_csv(data_url, sep="\s+", skiprows=22, header=None)
  CRIM      ZN  INDUS  CHAS    NOX     RM   AGE     DIS  RAD    TAX  \
0  0.00632  18.0    2.31   0.0  0.538  6.575   65.2  4.0900  1.0  296.0
1  0.02731   0.0    7.07   0.0  0.469  6.421   78.9  4.9671  2.0  242.0
2  0.02729   0.0    7.07   0.0  0.469  7.185   61.1  4.9671  2.0  242.0
3  0.03237   0.0    2.18   0.0  0.458  6.998   45.8  6.0622  3.0  222.0
4  0.06905   0.0    2.18   0.0  0.458  7.147   54.2  6.0622  3.0  222.0
5  0.02985   0.0    2.18   0.0  0.458  6.430   58.7  6.0622  3.0  222.0
6  0.08829  12.5    7.87   0.0  0.524  6.012   66.6  5.5605  5.0  311.0
7  0.14455  12.5    7.87   0.0  0.524  6.172   96.1  5.9505  5.0  311.0
8  0.21124  12.5    7.87   0.0  0.524  5.631  100.0  6.0821  5.0  311.0
9  0.17004  12.5    7.87   0.0  0.524  6.004   85.9  6.5921  5.0  311.0

  PTRATIO     B  LSTAT  MEDV
0    15.3  396.90   4.98  24.0
1    17.8  396.90   9.14  21.6
2    17.8  392.83   4.03  34.7
3    18.7  394.63   2.94  33.4
4    18.7  396.90   5.33  36.2
5    18.7  394.12   5.21  28.7
6    15.2  395.60  12.43  22.9
7    15.2  396.90  19.15  27.1
8    15.2  386.63  29.93  16.5
9    15.2  386.71  17.10  18.9
```

```
from sklearn.model_selection import train_test_split
```

```
X = df.loc[:, df.columns != 'MEDV']
y = df.loc[:, df.columns == 'MEDV']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=123)
```

```
from sklearn.preprocessing import MinMaxScaler
mms = MinMaxScaler()
mms.fit(X_train)
X_train = mms.transform(X_train)
X_test = mms.transform(X_test)
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

model = Sequential()

model.add(Dense(128, input_shape=(13, ), activation='relu', name='dense_1'))
model.add(Dense(64, activation='relu', name='dense_2'))
model.add(Dense(1, activation='linear', name='dense_output'))

model.compile(optimizer='adam', loss='mse', metrics=['mae'])
model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense_1 (Dense)	(None, 128)	1,792
dense_2 (Dense)	(None, 64)	8,256
dense_output (Dense)	(None, 1)	65

Total params: 10,113 (39.50 KB)

Trainable params: 10,113 (39.50 KB)

Non-trainable params: 0 (0.00 B)

```
history = model.fit(X_train, y_train, epochs=100, validation_split=0.05, verbose = 1)
```

```
Epoch 1/100
11/11 ----- 2s 24ms/step - loss: 570.0838 - mae: 22.0240 - val_loss: 578.3728 - val_mae: 22.0574
Epoch 2/100
11/11 ----- 0s 9ms/step - loss: 513.7112 - mae: 20.6480 - val_loss: 507.7814 - val_mae: 20.3472
Epoch 3/100
11/11 ----- 0s 9ms/step - loss: 431.6032 - mae: 18.5035 - val_loss: 405.7956 - val_mae: 17.5471
Epoch 4/100
11/11 ----- 0s 11ms/step - loss: 320.9745 - mae: 15.1837 - val_loss: 277.8407 - val_mae: 13.3673
Epoch 5/100
11/11 ----- 0s 9ms/step - loss: 203.3636 - mae: 11.2546 - val_loss: 168.6566 - val_mae: 9.0834
Epoch 6/100
11/11 ----- 0s 9ms/step - loss: 135.7076 - mae: 8.9652 - val_loss: 126.4323 - val_mae: 8.2509
Epoch 7/100
11/11 ----- 0s 9ms/step - loss: 119.6124 - mae: 8.4439 - val_loss: 116.9119 - val_mae: 7.9829
Epoch 8/100
11/11 ----- 0s 9ms/step - loss: 106.8513 - mae: 7.8536 - val_loss: 108.9030 - val_mae: 7.5646
Epoch 9/100
11/11 ----- 0s 8ms/step - loss: 93.8144 - mae: 7.2099 - val_loss: 101.8177 - val_mae: 7.2159
Epoch 10/100
11/11 ----- 0s 9ms/step - loss: 84.0362 - mae: 6.7147 - val_loss: 95.2713 - val_mae: 6.9154
Epoch 11/100
11/11 ----- 0s 9ms/step - loss: 75.9673 - mae: 6.3116 - val_loss: 89.7934 - val_mae: 6.6407
Epoch 12/100
11/11 ----- 0s 8ms/step - loss: 69.1915 - mae: 5.9932 - val_loss: 84.4670 - val_mae: 6.4754
Epoch 13/100
11/11 ----- 0s 10ms/step - loss: 63.9105 - mae: 5.7716 - val_loss: 80.1224 - val_mae: 6.4902
Epoch 14/100
11/11 ----- 0s 8ms/step - loss: 60.1558 - mae: 5.6003 - val_loss: 77.0768 - val_mae: 6.4176
Epoch 15/100
11/11 ----- 0s 10ms/step - loss: 56.6832 - mae: 5.3415 - val_loss: 75.4748 - val_mae: 6.2045
Epoch 16/100
11/11 ----- 0s 8ms/step - loss: 54.2835 - mae: 5.1663 - val_loss: 73.4424 - val_mae: 6.1518
Epoch 17/100
11/11 ----- 0s 10ms/step - loss: 52.1779 - mae: 5.0961 - val_loss: 71.2520 - val_mae: 6.1653
Epoch 18/100
11/11 ----- 0s 9ms/step - loss: 50.5135 - mae: 4.9874 - val_loss: 70.2099 - val_mae: 5.9805
Epoch 19/100
11/11 ----- 0s 8ms/step - loss: 48.7814 - mae: 4.9304 - val_loss: 67.1365 - val_mae: 6.1417
Epoch 20/100
11/11 ----- 0s 9ms/step - loss: 46.7747 - mae: 4.8462 - val_loss: 66.3817 - val_mae: 5.8444
Epoch 21/100
11/11 ----- 0s 11ms/step - loss: 44.9658 - mae: 4.6825 - val_loss: 64.5245 - val_mae: 5.7784
Epoch 22/100
11/11 ----- 0s 10ms/step - loss: 43.3153 - mae: 4.5780 - val_loss: 62.2418 - val_mae: 5.7626
Epoch 23/100
11/11 ----- 0s 10ms/step - loss: 41.7848 - mae: 4.6306 - val_loss: 59.7853 - val_mae: 5.7829
Epoch 24/100
11/11 ----- 0s 9ms/step - loss: 40.3323 - mae: 4.4149 - val_loss: 59.8869 - val_mae: 5.4106
Epoch 25/100
11/11 ----- 0s 9ms/step - loss: 37.8700 - mae: 4.2793 - val_loss: 56.6415 - val_mae: 5.5498
Epoch 26/100
11/11 ----- 0s 9ms/step - loss: 36.3189 - mae: 4.2472 - val_loss: 55.6554 - val_mae: 5.3065
Epoch 27/100
11/11 ----- 0s 9ms/step - loss: 34.5785 - mae: 4.0839 - val_loss: 54.0000 - val_mae: 5.2035
Epoch 28/100
11/11 ----- 0s 9ms/step - loss: 32.9806 - mae: 4.0004 - val_loss: 52.3326 - val_mae: 5.1136
Epoch 29/100
11/11 ----- 0s 13ms/step - loss: 31.2970 - mae: 3.8925 - val_loss: 50.6000 - val_mae: 5.0289
```

```
mse_nn, mae_nn = model.evaluate(X_test, y_test)
```

```
print('Mean squared error on test data: ', mse_nn)
```

```
print('Mean absolute error on test data: ', mae_nn)
```

```
5/5 ----- 0s 7ms/step - loss: 23.1358 - mae: 3.1223
```

```
Mean squared error on test data: 23.135820388793945
```

```
Mean absolute error on test data: 3.122297763824463
```